VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER ENGINEERING AND INFORMATION SYSTEM

DEPARTMENT: SOFTWARE ENGINEERING

NAME: VIGNESHWARAN T

REG NO: 23MIS0587

COURSE NAME: Agile Development Process and Devops Lab

COURSE CODE: ISWE406P

SLOT: L3 + L4

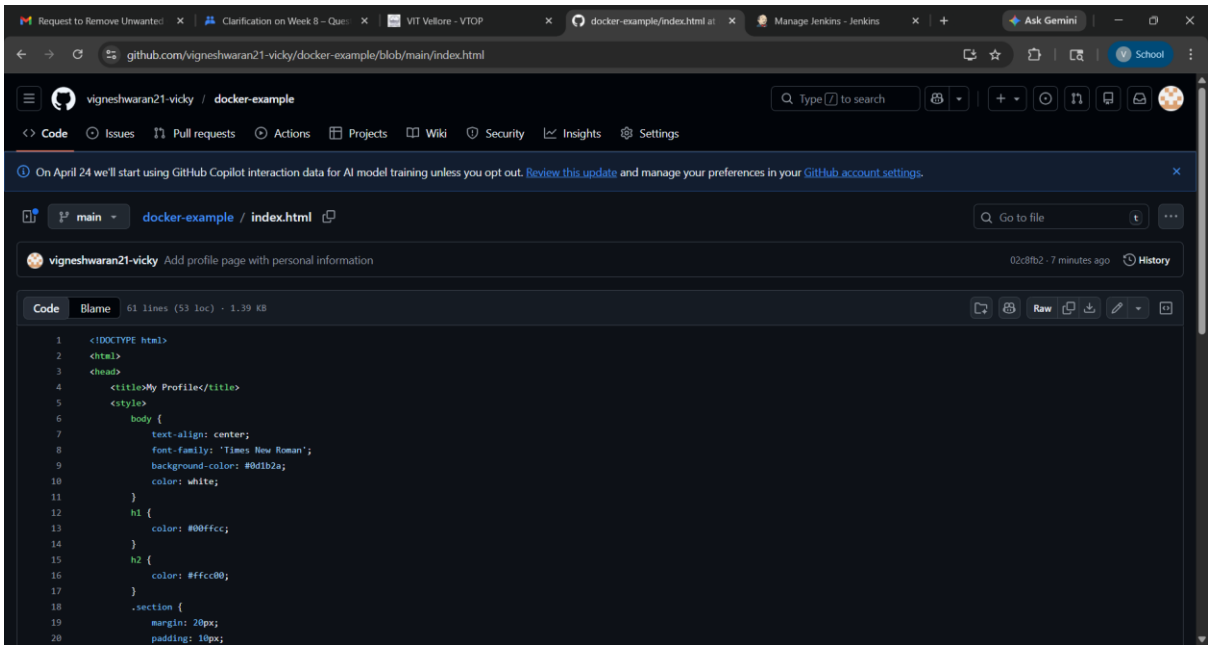
DIGITAL ASSIGNMENT – IV

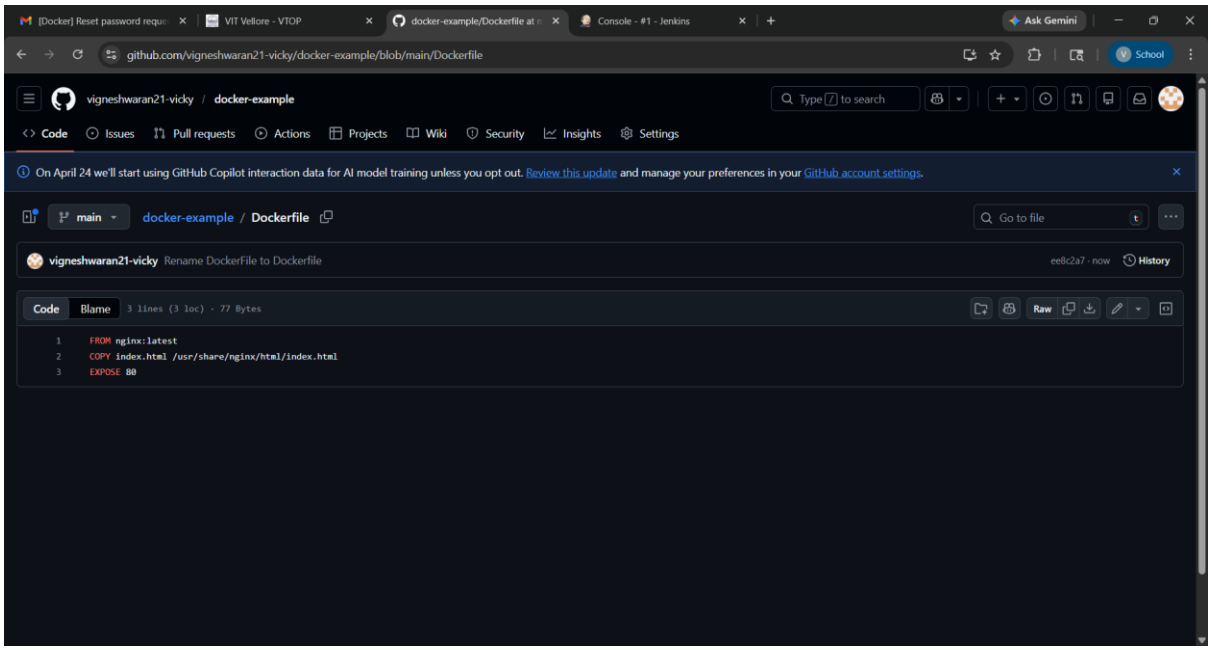
(Automated CI/CD Pipeline: Deploying a Web Application using Jenkins, Docker, and Kubernetes)

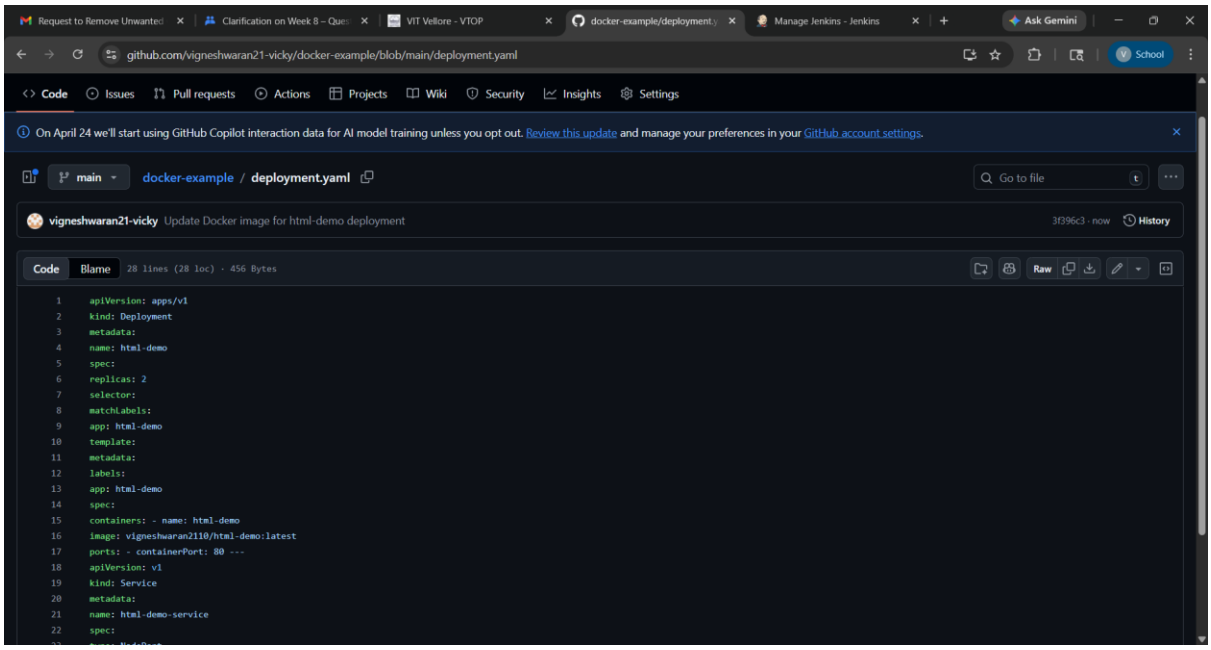
Create an HTML file for your profile and complete the following:

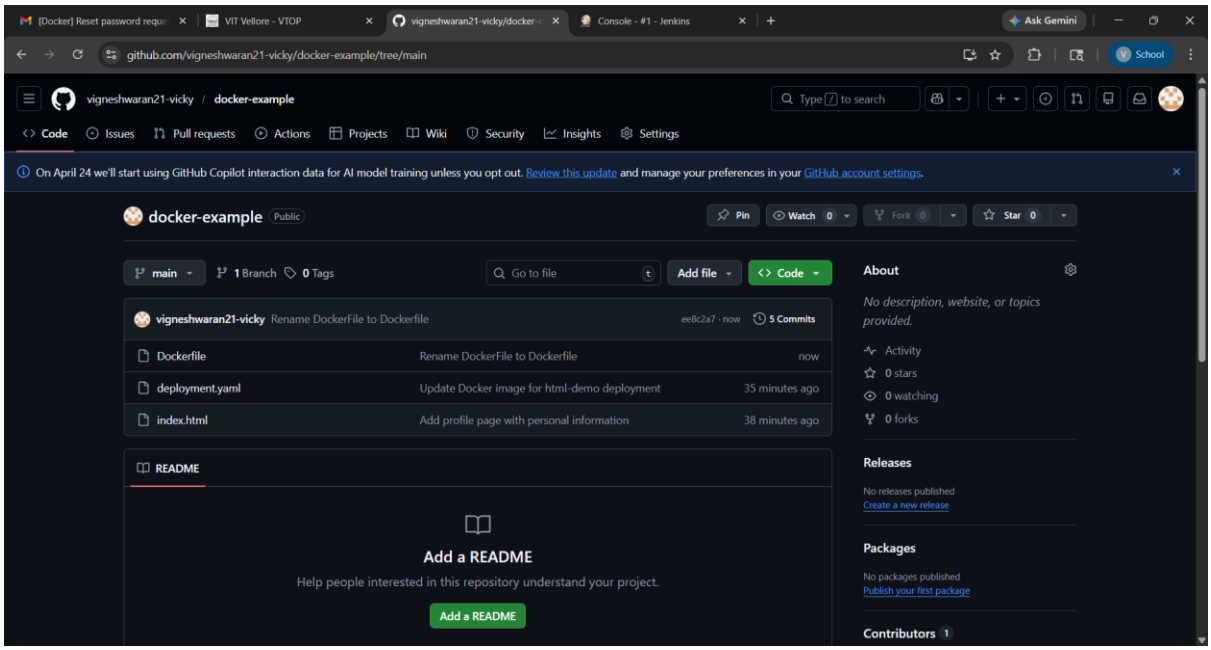
1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile**
- **Deployment.yaml**






Request to Remove Unwanted
Clarification on Week 8 – Ques
VIT Vellore - VTOP
docker-example/deployment.y
Manage Jenkins - Jenkins
Ask Gemini
github.com/vigneshwaran21-vicky/docker-example/blob/main/deployment.yaml
School
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
main
docker-example / deployment.yaml
Go to file
vigneshwaran21-vicky Update Docker image for html-demo deployment
3f396c3 · now
History
Code
Blame
28 lines (28 loc) · 456 Bytes
Raw
apiVersion: apps/v1
kind: Deployment
metadata:
name: html-demo
spec:
replicas: 2
selector:
matchLabels:
app: html-demo
template:
metadata:
labels:
app: html-demo
spec:
containers: - name: html-demo
image: vigneshwaran2110/html-demo:latest
ports: - containerPort: 80 ---
apiVersion: v1
kind: Service
metadata:
name: html-demo-service
spec:


[Docker] Reset password reque
VIT Vellore - VTOP
vigneshwaran21-vicky/docker-
Console - #1 - Jenkins
Ask Gemini
github.com/vigneshwaran21-vicky/docker-example/tree/main
School
vigneshwaran21-vicky / docker-example
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
docker-example Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
vigneshwaran21-vicky Rename DockerFile to Dockerfile
ee8c2a7 · now
5 Commits
Dockerfile
Rename DockerFile to Dockerfile
now
deployment.yaml
Update Docker image for html-demo deployment
35 minutes ago
index.html
Add profile page with personal information
38 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1

The code for the three files are given below:
Create index.html:

```
<!DOCTYPE html>
<html>
<head>
   <title>My Profile</title>
   <style>
      body {
         text-align: center;
         font-family: 'Times New Roman';
         background-color: #0d1b2a;
         color: white;
      }
      h1 {
         color: #00ffcc;
      }
      h2 {
         color: #ffcc00;
      }
      .section {
         margin: 20px;
         padding: 10px;
      }
   </style>
</head>

<body>

   <h1>My Profile</h1>
   <h2>Vigneshwaran T</h2>

   <div class="section">
      <h3>Education</h3>
      <p>M.Tech Integrated Software Engineering</p>
   </div>

   <div class="section">
      <h3>About Me</h3>
      <p>
         I am a passionate software engineering student with strong interest in
         development, data science, and innovative technologies. I enjoy building
         real-world applications and continuously learning new skills.
      </p>
   </div>

   <div class="section">
      <h3>Achievements</h3>
      <p>✓ Published 2 Patents</p>
```

```
        <p>✓ Participated in technical projects and research activities</p>
    </div>

    <div class="section">
        <h3>Skills</h3>
        <p>Python, Machine Learning, Web Development, Data Analysis</p>
    </div>

    <div class="section">
        <h3>Contact</h3>
        <p>Email: vigneshwaran.2023b@vitstudent.ac.in</p>
    </div>

</body>
</html>
```

Create a Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

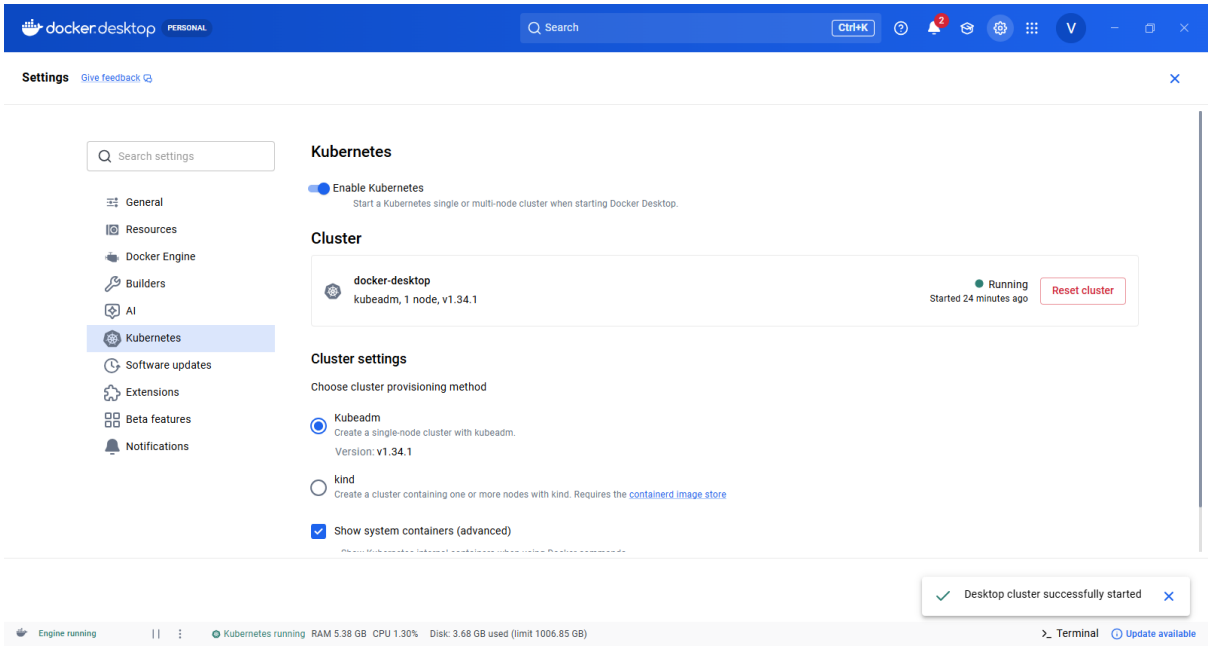
deployment.yaml:

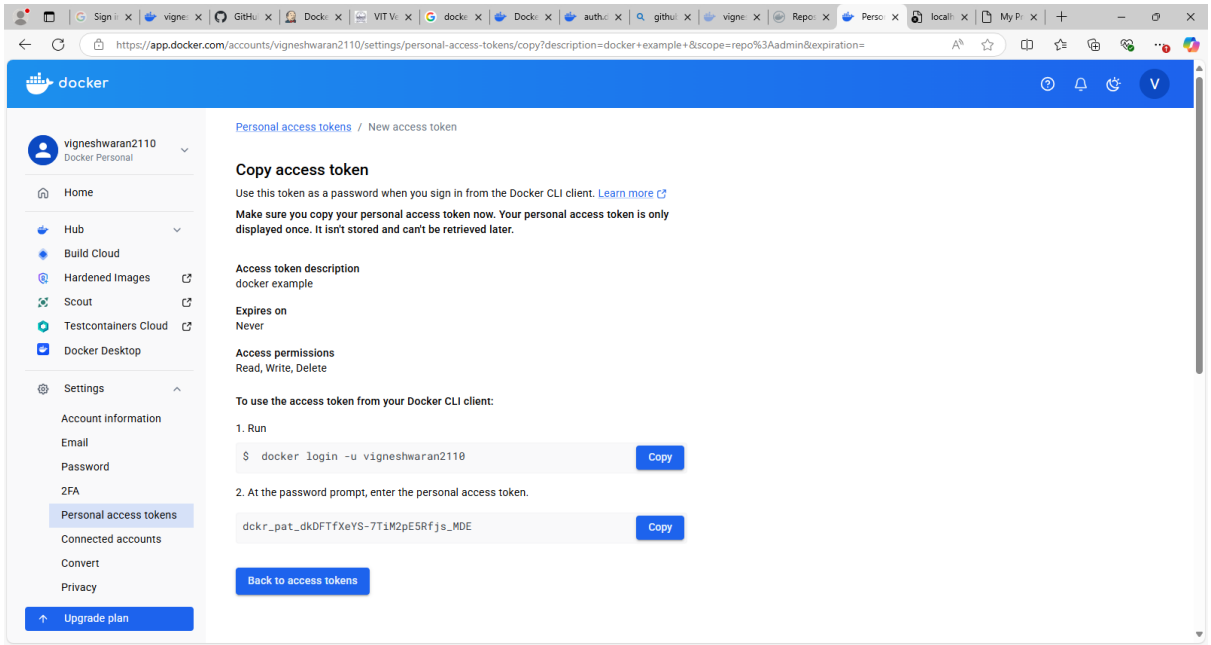
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
        - name: html-demo
          image: vigneshwaran2110/html-demo:latest
          ports:
            - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
```

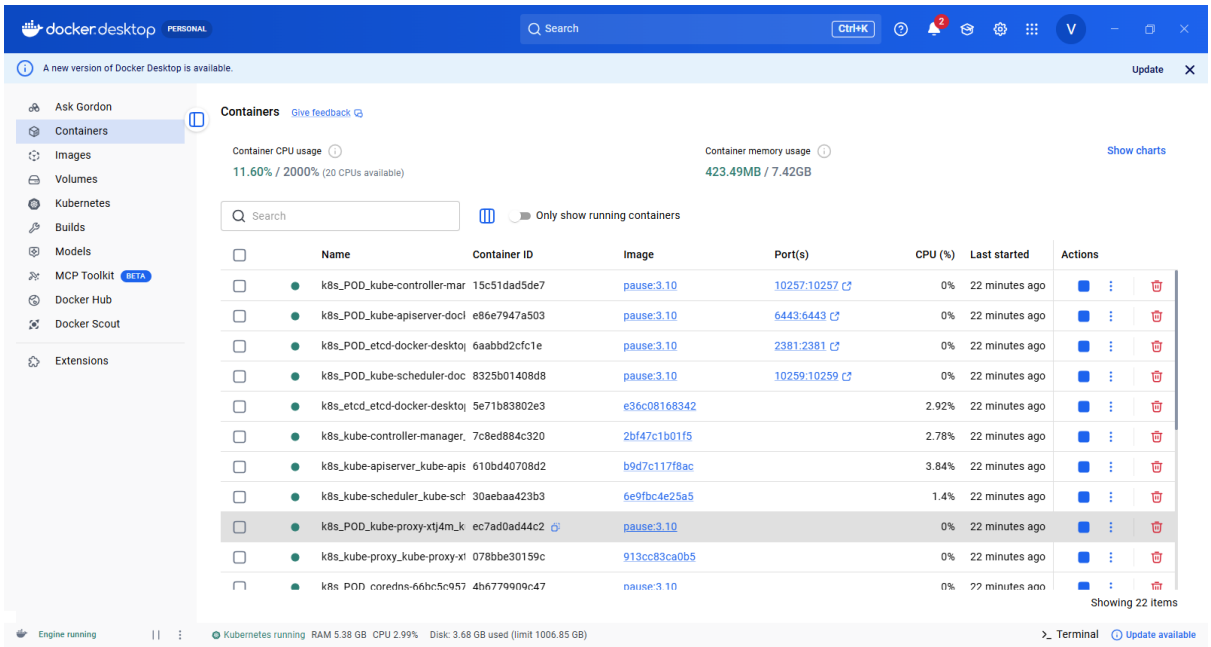
```
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010
```

2. Enable Kubernetes

- ✓ **Open the Docker Dashboard:** Launch the Docker Desktop application on your system.
- ✓ **Navigate to Settings:** Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.
- ✓ **Select the Kubernetes Tab:** In the Settings menu, find and select the **Kubernetes** tab.
- ✓ **Enable Kubernetes:** Check the box labeled **Enable Kubernetes.**
- ✓ **Apply & Restart:** Click the **Apply & Restart button** at the bottom of the settings window.
- ✓ Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time

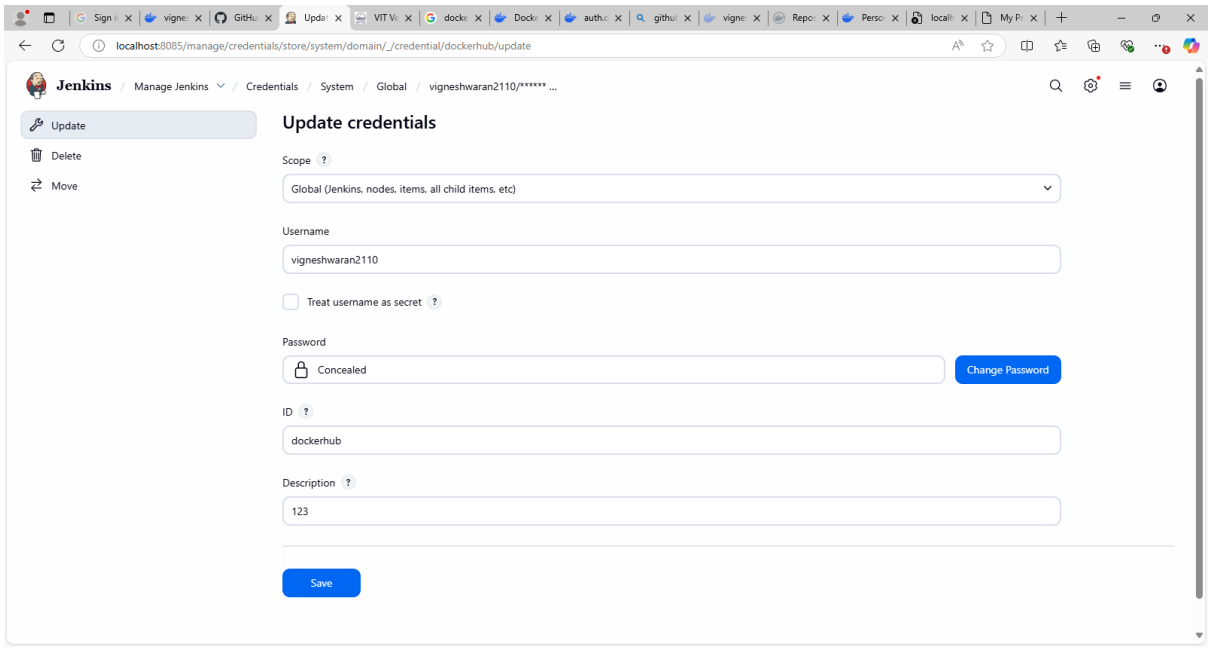



https://app.docker.com/accounts/vigneshwaran2110/settings/personal-access-tokens/copy?description=docker+example+&scope=repo%3Aadmin&expiration=
docker
vigneshwaran2110
Docker Personal
Home
Hub
Build Cloud
Hardened Images
Scout
Testcontainers Cloud
Docker Desktop
Settings
Account information
Email
Password
2FA
Personal access tokens
Connected accounts
Convert
Privacy
Upgrade plan
Personal access tokens / New access token
Copy access token
Use this token as a password when you sign in from the Docker CLI client. Learn more
Make sure you copy your personal access token now. Your personal access token is only displayed once. It isn't stored and can't be retrieved later.
Access token description
docker example
Expires on
Never
Access permissions
Read, Write, Delete
To use the access token from your Docker CLI client:
1. Run
$ docker login -u vigneshwaran2110
Copy
2. At the password prompt, enter the personal access token.
dckr_pat_dkDFTfXeYS-7TiM2pE5Rfjs_MDE
Copy
Back to access tokens

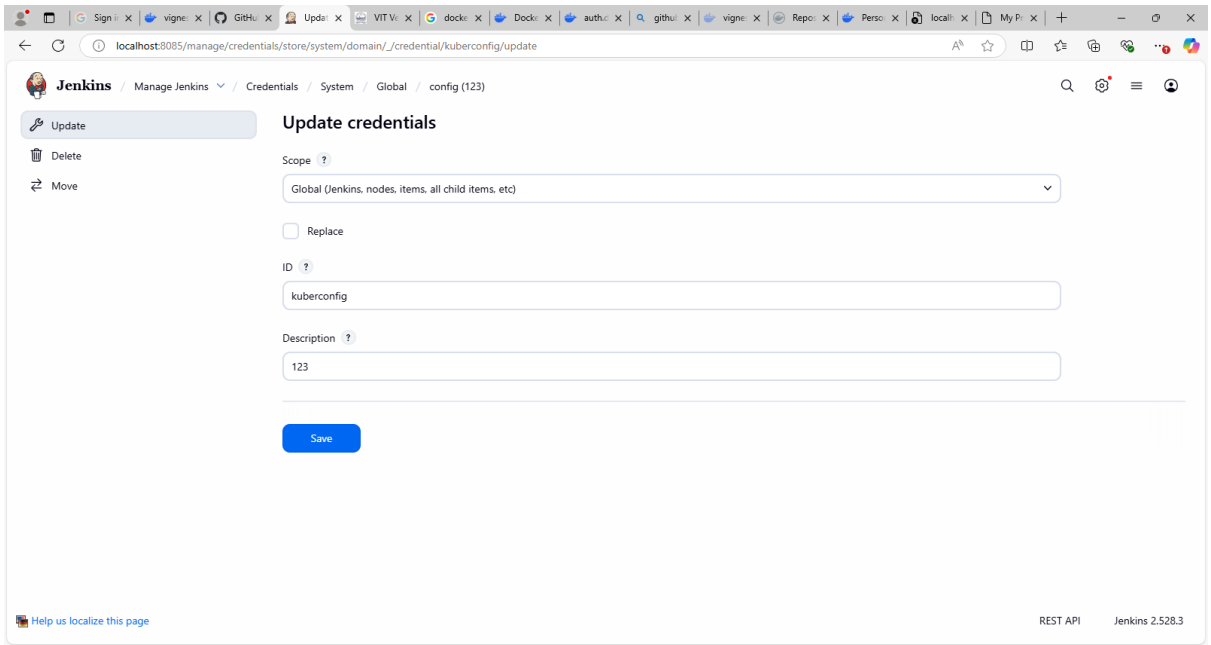

docker desktop PERSONAL
Search
Ctrl+K
A new version of Docker Desktop is available.
Update
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
Container CPU usage
11.60% / 2000% (20 CPUs available)
Container memory usage
423.49MB / 7.42GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
k8s_POD_kube-controller-mar 15c51dad5de7 pause:3.10 10257:10257 0% 22 minutes ago
k8s_POD_kube-apiserver-docl e86e7947a503 pause:3.10 6443:6443 0% 22 minutes ago
k8s_POD_etcd-docker-deskto 6aabbd2cfc1e pause:3.10 2381:2381 0% 22 minutes ago
k8s_POD_kube-scheduler-doc 8325b01408d8 pause:3.10 10259:10259 0% 22 minutes ago
k8s_etcd_etcd-docker-deskto 5e71b83802e3 e36c08168342 2.92% 22 minutes ago
k8s_kube-controller-manager_ 7c8ed884c320 2bf47c1b01f5 2.78% 22 minutes ago
k8s_kube-apiserver_kube-apis 610bd40708d2 b9d7c117f8ac 3.84% 22 minutes ago
k8s_kube-scheduler_kube-sch 30aebaa423b3 6e9fbc4e25a5 1.4% 22 minutes ago
k8s_POD_kube-proxy-xtj4m_k ec7ad0ad44c2 pause:3.10 0% 22 minutes ago
k8s_kube-proxy_kube-proxy-xt 078bbe30159c 913cc83ca0b5 0% 22 minutes ago
Showing 22 items
Engine running
Kubernetes running RAM 5.38 GB CPU 2.99% Disk: 3.68 GB used (limit 1006.85 GB)
Terminal
Update available

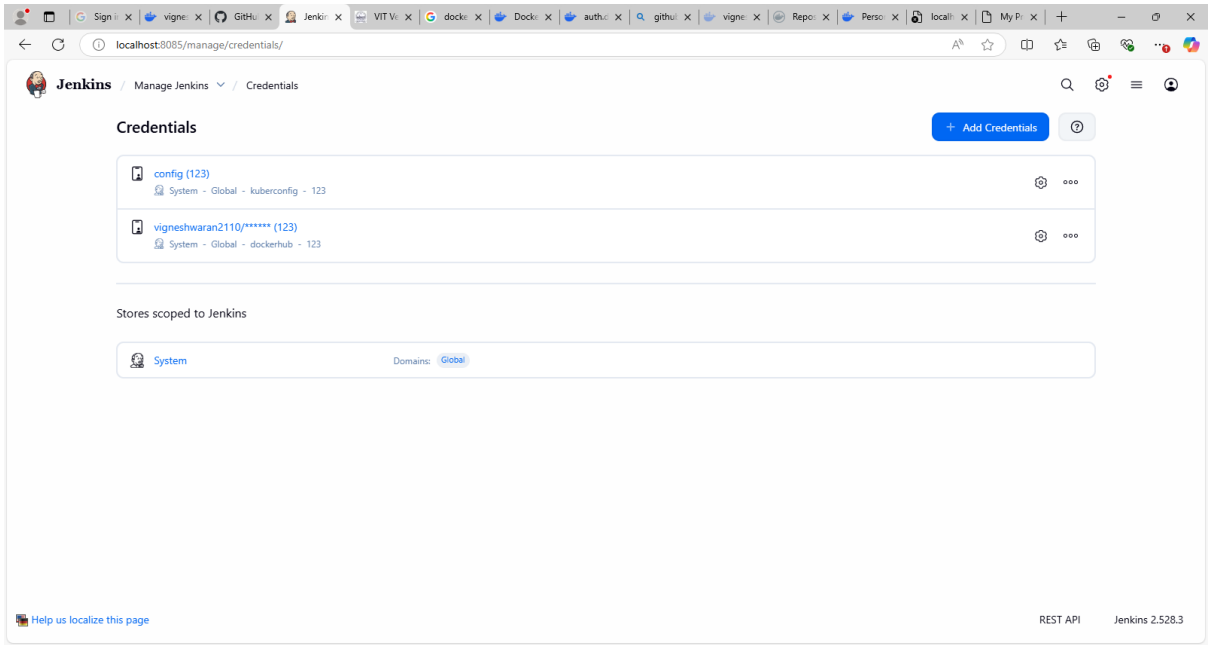
3. Open Jenkins In Jenkins

Setting --> Credentials New Credentials – User name: , password , ID:, Description



New Credentials – Secret – Broswse C: User->Administrator-->.kube





4. Create New Item -> Pipeline
Paste the code in pipeline script

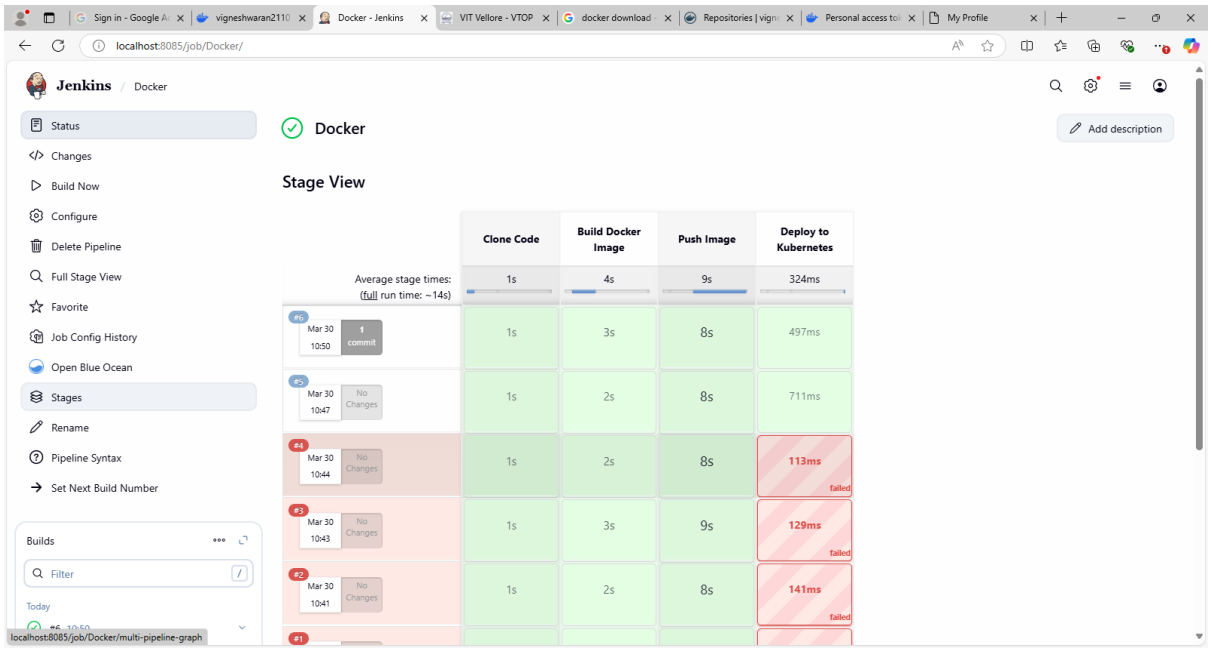
- ✓ Update URL of GIT repository
- ✓ **Make sure credentialsId: "dockerhub"** same as mentioned in credentials
- ✓ Make sure that the ID mentioned in credentials "**kuberconfig**" is same as in script
- ✓ Run the Build

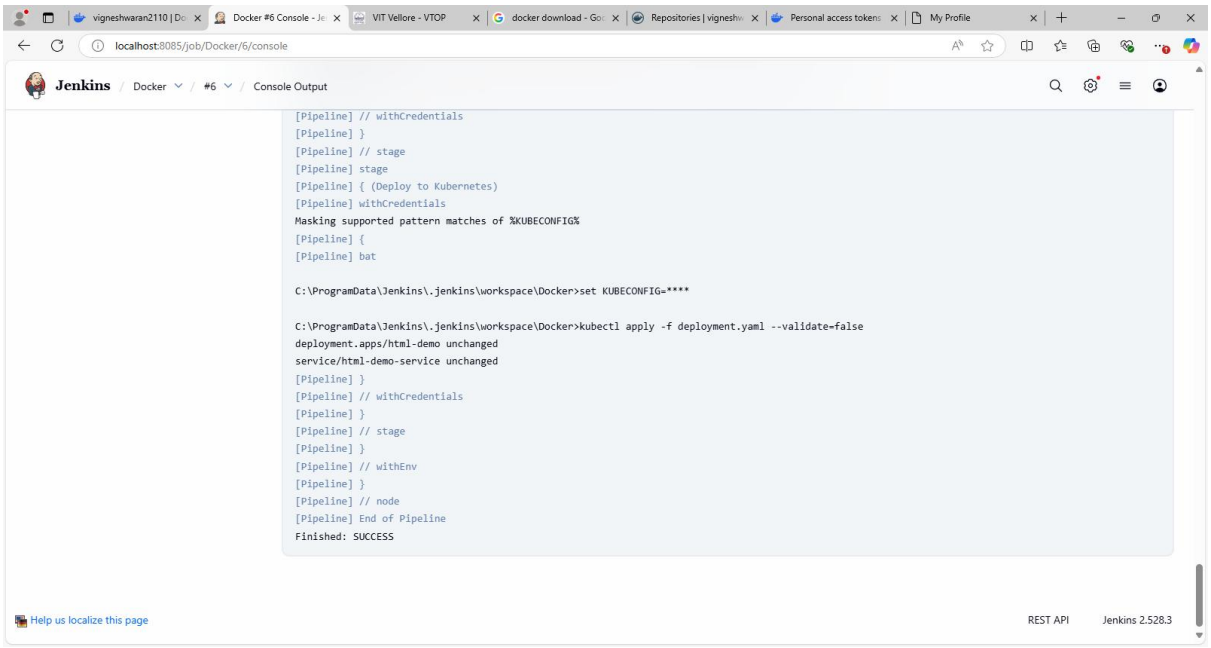
<u>Pipeline:</u>

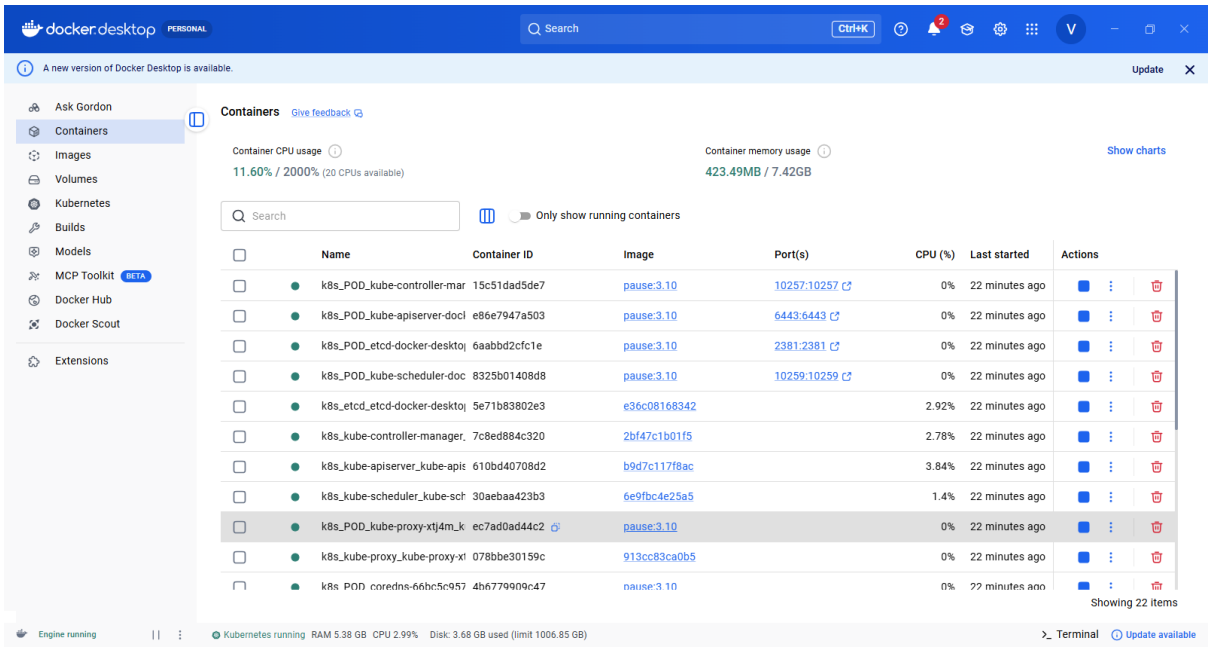
```
pipeline {
agent any
environment {
DOCKER_IMAGE = "vigneshwaran2110/html-demo"
}
stages {
 stage('Clone Code') {
steps {
git branch: 'main',
url: 'https://github.com/vigneshwaran21-vicky/docker-example.git'
}
}
stage('Build Docker Image') {
steps {
bat 'docker build -t %DOCKER_IMAGE% .'
}
```

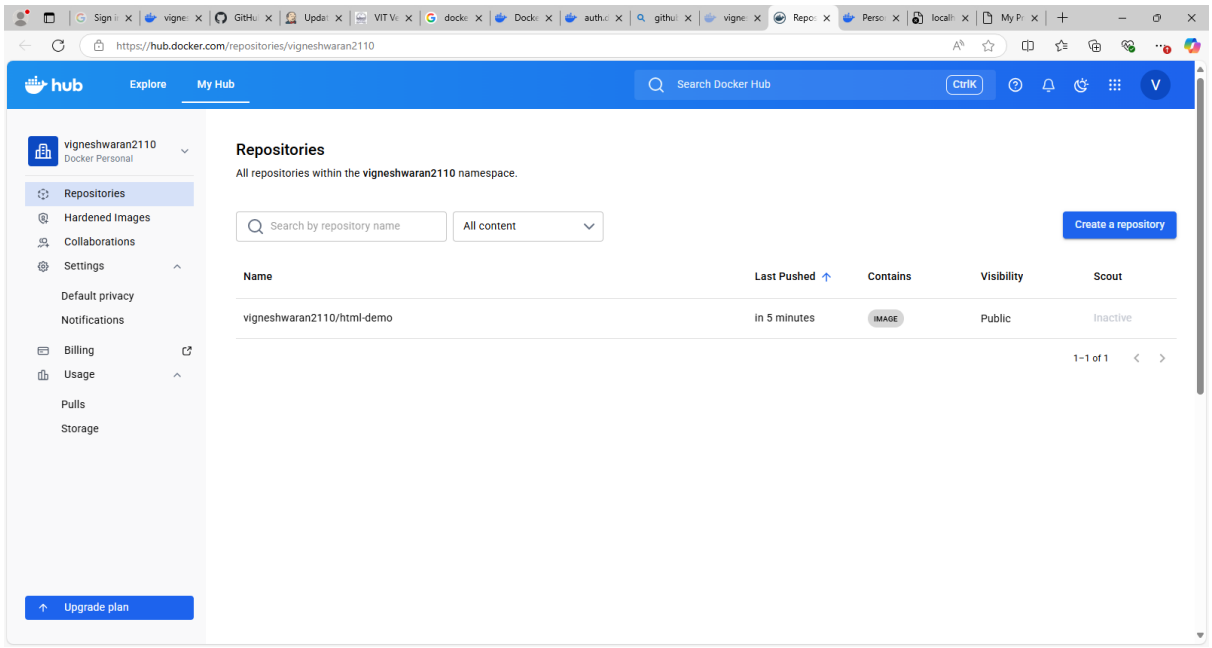
```
}
stage('Push Image') {
steps {
withCredentials([usernamePassword(credentialsId: 'dockerhub',
usernameVariable: 'USER', passwordVariable: 'PASS')]) {
bat 'docker login -u %USER% -p %PASS%'
bat 'docker push %DOCKER_IMAGE%'
}
}
}
stage('Deploy to Kubernetes') {
steps {
withCredentials([file(credentialsId: 'kuberconfig', variable:
'KUBECONFIG')]) {
bat '''
set KUBECONFIG=%KUBECONFIG%
kubectl apply -f deployment.yaml --validate=false
'''
}
}
}
}
}
```

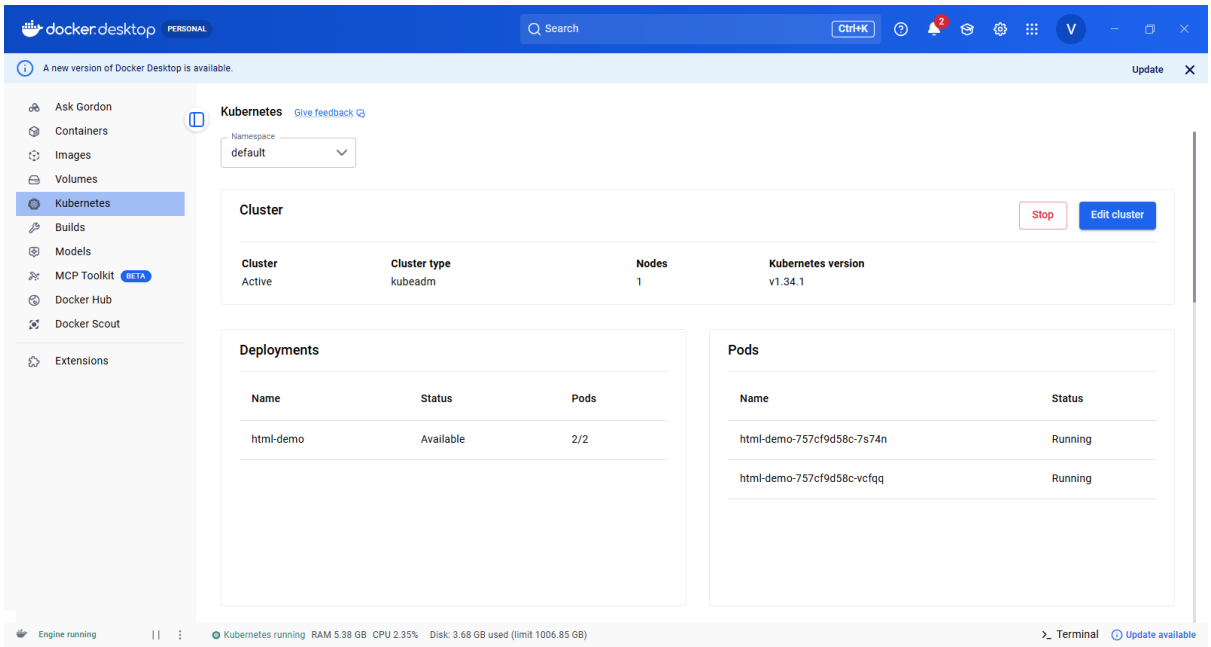
OTUPUT:-

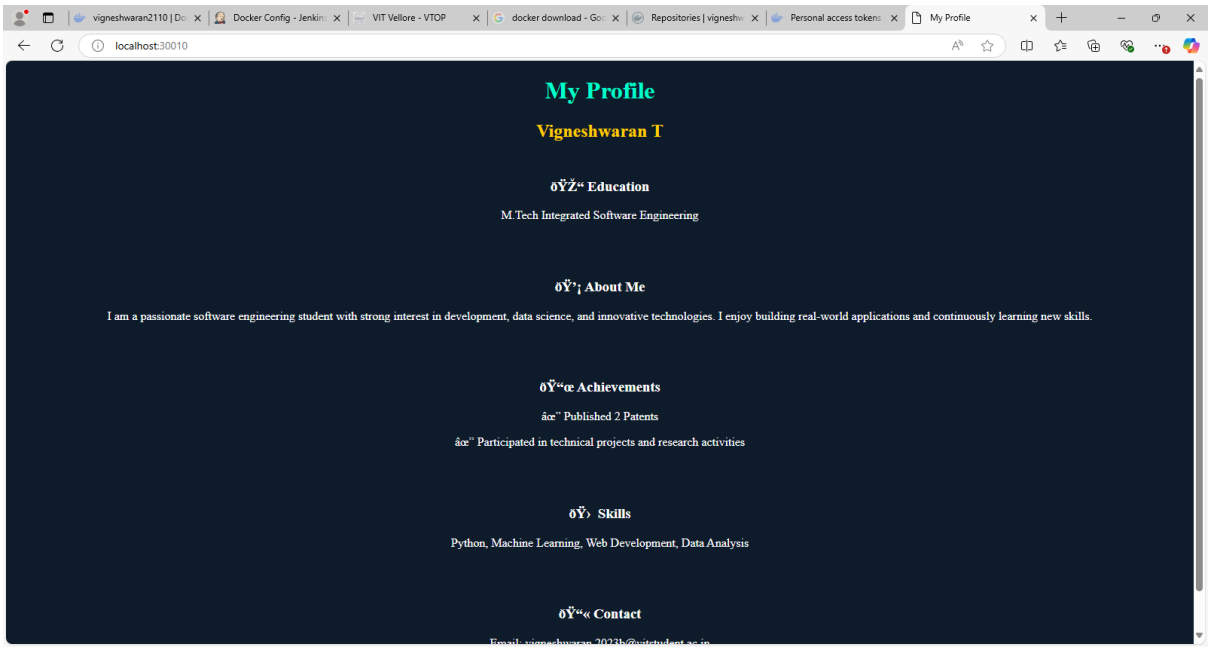



Docker #6 Console
localhost:8085/job/Docker/6/console
Jenkins / Docker / #6 / Console Output
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\Docker>set KUBECONFIG=****
C:\ProgramData\Jenkins\.jenkins\workspace\Docker>kubectl apply -f deployment.yaml --validate=false
deployment.apps/html-demo unchanged
service/html-demo-service unchanged
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Help us localize this page
REST API
Jenkins 2.528.3


docker desktop PERSONAL
Search Ctrl+K
A new version of Docker Desktop is available.
Update
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
Container CPU usage
11.60% / 2000% (20 CPUs available)
Container memory usage
423.49MB / 7.42GB
Show charts
Search
Only show running containers
Name Container ID Image Port(s) CPU (%) Last started Actions
k8s_POD_kube-controller-mar 15c51dad5de7 pause:3.10 10257:10257 0% 22 minutes ago
k8s_POD_kube-apiserver-docl e86e7947a503 pause:3.10 6443:6443 0% 22 minutes ago
k8s_POD_etcd-docker-desktoj 6aabbd2cfc1e pause:3.10 2381:2381 0% 22 minutes ago
k8s_POD_kube-scheduler-doc 8325b01408d8 pause:3.10 10259:10259 0% 22 minutes ago
k8s_etcd_etcd-docker-desktoj 5e71b83802e3 e36c08168342 2.92% 22 minutes ago
k8s_kube-controller-manager_ 7c8ed884c320 2bf47c1b01f5 2.78% 22 minutes ago
k8s_kube-apiserver_kube-apis 610bd40708d2 b9d7c117f8ac 3.84% 22 minutes ago
k8s_kube-scheduler_kube-sch 30aebaa423b3 6e9fbc4e25a5 1.4% 22 minutes ago
k8s_POD_kube-proxy-xtj4m_k ec7ad0ad44c2 pause:3.10 0% 22 minutes ago
k8s_kube-proxy_kube-proxy-xt 078bbe30159c 913cc83ca0b5 0% 22 minutes ago
Showing 22 items
Engine running
Kubernetes running RAM 5.38 GB CPU 2.99% Disk: 3.68 GB used (limit 1006.85 GB)
Terminal
Update available


https://hub.docker.com/repositories/vigneshwaran2110
hub
Explore
My Hub
Search Docker Hub
Ctrl K
vigneshwaran2110
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
Repositories
All repositories within the vigneshwaran2110 namespace.
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
vigneshwaran2110/html-demo
in 5 minutes
IMAGE
Public
Inactive
1–1 of 1


docker desktop
PERSONAL
Search
Ctrl+K
A new version of Docker Desktop is available.
Update
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Docker Scout
Extensions
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kubeadm
Nodes
1
Kubernetes version
v1.34.1
Deployments
Name
Status
Pods
html-demo
Available
2/2
Pods
Name
Status
html-demo-757cf9d58c-7s74n
Running
html-demo-757cf9d58c-vcfqq
Running
Engine running
Kubernetes running
RAM 5.38 GB
CPU 2.35%
Disk: 3.68 GB used (limit 1006.85 GB)
Terminal
Update available


vigneshwaran2110 | Do
Docker Config - Jenkin
VIT Vellore - VTOP
docker download - Go
Repositories | vignesh
Personal access tokens
My Profile
localhost:30010
My Profile
Vigneshwaran T
ðŸŽ“ Education
M.Tech Integrated Software Engineering
ðŸ'¡ About Me
I am a passionate software engineering student with strong interest in development, data science, and innovative technologies. I enjoy building real-world applications and continuously learning new skills.
ðŸ“œ Achievements
âœ" Published 2 Patents
âœ" Participated in technical projects and research activities
ðŸ› Skills
Python, Machine Learning, Web Development, Data Analysis
ðŸ“« Contact